ECE 562: Deep Reinforcement Learning for Robotics

Fall 2026

Assignment 3: PPO II — Stale Data and Off-Policy Corrections

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PPO is often described as "almost on-policy". After every PPO iteration it discards its rollout and collects a new one with the current policy, even though its surrogate objective contains an importance ratio, which is an off-policy tool. This assignment has two goals:

- to understand why PPO is still an on-policy method, and the limitations of the off-policy policy gradient estimator it uses; and
- to understand how PPO's clipped objective helps, and where it does not.

The assignment is structured into two main sections:

- **Theory (40 points):** Trajectory- and state-level importance sampling, the missing state-distribution correction in PPO, the fully corrected off-policy policy gradient, and why the state-distribution ratio is hard to compute.
- **Experiments and Report (60 points):** You are given a complete, working PPO implementation for `Hopper-v5`. You will modify it to train on stale data and compare three surrogate losses.

To earn full credit, clearly define all notation and assumptions, show the important intermediate steps when a derivation is requested, and provide clear justifications for each empirical finding. If a result does not match what you expected, report it and explain what might cause it.

Unless otherwise stated, use the following assumptions:

- For the derivations in Section 1, the state space $\mathcal{S}$ and action space $\mathcal{A}$ are finite and $\sum$ denotes summation; for continuous spaces, replace sums by integrals.
- Rewards are bounded, and the discount factor satisfies $0 \leq \gamma < 1$.
- $\pi_{\theta_{\text{old}}}$ denotes the *behavior* policy that generated the data, and π_θ the policy being optimized. We assume $\pi_{\theta_{\text{old}}}(a \mid s) > 0$ whenever $\pi_\theta(a \mid s) > 0$ (this holds for the Gaussian policies used in this course), and $d^{\pi_{\theta_{\text{old}}}}(s) > 0$ whenever $d^{\pi_\theta}(s) > 0$.
- p_0 is the initial state distribution and $P(s' \mid s, a)$ the transition kernel. The normalized discounted state visitation distribution of a policy π is

$$d^\pi(s) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \, \mathbb{P}(s_t = s \mid s_0 \sim p_0,\ \pi).$$

- With this normalization, the policy gradient theorem reads

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^{\pi_\theta},\ a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a \mid s)\, A^{\pi_\theta}(s,a)\right].$$

- The per-step importance ratio is

$$r_\theta(s,a) = \frac{\pi_\theta(a \mid s)}{\pi_{\theta_{\text{old}}}(a \mid s)}.$$

Submission Instructions Submit your completed assignment through Canvas containing:

- **`report.pdf`:** Your solutions to the theoretical questions in Section 1, and your description of your code modifications, plots, and written analysis for Section 2. You may prepare this report in LaTeX or write it neatly by hand and scan it into a clear PDF.
- **`code/`:** Your modified codebase, together with a short `README.md` listing the exact commands that produce every figure in your report. Your implementation must run without coding errors.

1 Theory (40 points)

This section examines what PPO's importance ratio corrects, what it leaves uncorrected, and why PPO must keep its data (almost) on-policy.

Q1 **(8 points) Trajectory-Level Importance Sampling**

Consider trajectories

$$\tau = (s_0, a_0, s_1, a_1, \ldots, s_{T-1}, a_{T-1}, s_T)$$

of a fixed length T. For this question only, assume that the reward r_t is determined by (s_t, a_t, s_{t+1}). Let

$$R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t$$

be its discounted return, and define the finite-horizon objective

$$J_T(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}[R(\tau)] = \sum_\tau p_\theta(\tau) R(\tau).$$

The trajectory probability under π_θ is

$$p_\theta(\tau) = p_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t \mid s_t) P(s_{t+1} \mid s_t, a_t).$$

(a) **(3 points)** Starting from the definition of $J_T(\theta)$, show that

$$J_T(\theta) = \mathbb{E}_{\tau\sim\pi_{\theta_{\text{old}}}}\left[\frac{p_\theta(\tau)}{p_{\text{old}}(\tau)} R(\tau)\right].$$

State exactly where the support assumption is used.

(b) **(3 points)** Using the trajectory probability above, show that the trajectory ratio does not depend on the dynamics:

$$\frac{p_\theta(\tau)}{p_{\text{old}}(\tau)} = \prod_{t=0}^{T-1} \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}.$$

Why does this matter for model-free reinforcement learning?

(c) **(2 points)** Explain why this estimator becomes impractical for long horizons, and why PPO instead uses the single-step ratio $r_\theta(s_t, a_t)$.

Q2 **(12 points) The Missing State-Distribution Correction**

Suppose samples are generated by the behavior policy:

$$s \sim d^{\pi_{\theta_{\text{old}}}}, \qquad a \sim \pi_{\theta_{\text{old}}}(a \mid s).$$

The distribution we actually want to sample from when optimizing π_θ is

$$d^{\pi_\theta}(s)\, \pi_\theta(a \mid s).$$

The full state-action distribution ratio can therefore be written as

$$\frac{d^{\pi_\theta}(s)\, \pi_\theta(a \mid s)}{d^{\pi_{\theta_{\text{old}}}}(s)\, \pi_{\theta_{\text{old}}}(a \mid s)} = w_\theta(s)\, r_\theta(s, a), \qquad w_\theta(s) = \frac{d^{\pi_\theta}(s)}{d^{\pi_{\theta_{\text{old}}}}(s)}.$$

(a) **(5 points)** Use the expression above to explain why the single-step ratio $r_\theta(s, a)$ corrects the conditional action distribution but not the state visitation distribution. Why should the missing state-distribution correction become more important when the data was generated by an increasingly stale policy?

(b) **(7 points)** Define the total variation distance as

$$D_{\text{TV}}(p, q) = \frac{1}{2} \sum_a |p(a) - q(a)|.$$

The following bound is known (Achiam et al., 2017):

$$\|d^{\pi_\theta} - d^{\pi_{\theta_{\text{old}}}}\|_1 \leq \frac{2\gamma}{1-\gamma}\, \mathbb{E}_{s \sim d^{\pi_{\theta_{\text{old}}}}} \left[D_{\text{TV}} \left(\pi_\theta(\cdot \mid s), \pi_{\theta_{\text{old}}}(\cdot \mid s) \right) \right].$$

Use it to explain why keeping π_θ close to $\pi_{\theta_{\text{old}}}$ helps limit the overall state-visitation mismatch, and how PPO's clipped objective helps keep the policy update small.

Does clipping strictly enforce a bound on D_{TV}?

The bound above can be loose in practice; interpret it as a qualitative motivation rather than a tight numerical guarantee.

Q3 **(8 points) The Fully Corrected Off-Policy Policy Gradient**

Now suppose the two policies are no longer close.

(a) **(4 points)** Using the policy gradient theorem and the state-action distribution ratio in Q2, show that

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma}\, \mathbb{E}_{s \sim d^{\pi_{\theta_{\text{old}}}},\ a \sim \pi_{\theta_{\text{old}}}} \left[w_\theta(s)\, r_\theta(s, a)\, \nabla_\theta \log \pi_\theta(a \mid s)\, A^{\pi_\theta}(s, a) \right].$$

(b) **(4 points)** The gradient PPO actually uses (before clipping) is

$$\hat{\mathbb{E}}_{(s,a)\sim\pi_{\theta_{\text{old}}}}\left[r_\theta(s,a)\,\nabla_\theta \log \pi_\theta(a \mid s)\,\hat{A}(s,a)\right],$$

where $\hat{A}$ is computed with GAE on trajectories generated by $\pi_{\theta_{\text{old}}}$.
To isolate the two main off-policy approximations, assume for this comparison that the sampled states follow $d^{\pi_{\theta_{\text{old}}}}$ and that

$$\hat{A}(s,a) = A^{\pi_{\theta_{\text{old}}}}(s,a).$$

Ignoring constant factors, identify the two approximations PPO makes relative to the expression in part (a):

$$w_\theta(s) \approx 1, \qquad A^{\pi_\theta}(s,a) \approx A^{\pi_{\theta_{\text{old}}}}(s,a).$$

Show that both approximations become exact at $\theta = \theta_{\text{old}}$. In practical PPO, why is GAE with a learned critic still only an estimate of $A^{\pi_{\theta_{\text{old}}}}(s,a)$?

Q4 **(10 points) Why Is $w_\theta(s)$ Hard to Compute?**

The normalized discounted state visitation distribution satisfies

$$d^\pi(s') = (1-\gamma)\,p_0(s') + \gamma \sum_s d^\pi(s) \sum_a \pi(a \mid s)\,P(s' \mid s,a).$$

Using

$$P_\pi(s,s') = \sum_a \pi(a \mid s) P(s' \mid s,a),$$

the same equation can be written in matrix form as

$$d^\pi = (1-\gamma)\left(I - \gamma P_\pi^\top\right)^{-1} p_0.$$

(a) **(6 points)** Which terms in these equations make

$$w_\theta(s) = \frac{d^{\pi_\theta}(s)}{d^{\pi_{\theta_{\text{old}}}}(s)}$$

hard to compute in model-free reinforcement learning with continuous states?

(b) **(4 points)** Explain why a change of π_θ in one state can change d^{π_θ} in many other states. Without the state-distribution correction $w_\theta(s)$, what happens to the quality of a PPO update when the data was generated by a policy from many PPO iterations ago? Connect your answer to Q2.

2 Experiments and Report (60 points)

Unlike Assignment 2, there are no `TODO` sections. You are given a complete PPO implementation for `Hopper-v5` (3-dimensional continuous actions); running `python train.py` trains standard PPO. It consists of:

- `actor_critic.py`: Gaussian actor with a state-independent $\log \sigma$ and a state-value critic (2 hidden layers of 64 units, tanh).

- `buffer.py`: Rollout storage and GAE. Each rollout stores the behavior log-probabilities $\log \pi_{\theta_{\text{old}}}(a_t \mid s_t)$, the values $V(s_t)$ at collection time, and the bootstrap state s_T with its value.
- `ppo.py`: The PPO mini-batch update (clipped surrogate, value loss, gradient clipping).
- `train.py`: Rollout collection, observation and reward normalization, evaluation, and CSV logging. The function `ratio_stats` logs, for the batch used in each update, the approximate KL divergence between the policy that collected the batch and the current policy, percentiles of r_θ, and the clip fraction, both before (`_start`) and after (`_end`) the update.

You will modify this code. You may structure your changes however you like (for example, by adding command-line flags), but you must implement each modification exactly as specified below so that results are comparable across students. Do not change the default hyperparameters:

Rollout length	2048	Discount γ	0.99
Mini-batch size	64	GAE λ	0.95
Epochs per PPO iteration	10	Clip ϵ	0.2
Learning rate	3×10^{-4}	Entropy coefficient	0
Value loss coefficient	0.5	Gradient clipping	0.5

Run every configuration for 3×10^5 environment steps with seeds $\{1, 2, 3\}$. Every learning curve must show the mean across seeds with a shaded region of $\pm$ one standard deviation, with environment steps on the x-axis. A run takes roughly 5–15 minutes on one CPU core; runs are independent and can be executed in parallel.

Throughout, one *PPO iteration* means collecting one rollout followed by the 10 epochs of mini-batch optimization above. The *data lag* k of a rollout is the number of PPO iterations between the policy that collected it and the current policy: a rollout with lag k was collected by $\pi_{\theta_{i-k}}$ and is used at PPO iteration i.

Q5 **(30 points) How Stale Can PPO's Data Be?**

(a) **(10 points) Modification.** Add a staleness parameter k. At every PPO iteration i, still collect one fresh rollout with the current policy (so every run uses the same number of environment steps), but perform the update on the rollout collected at iteration $i-k$. Keep the $k+1$ most recent rollouts; early in training, when fewer than $k+1$ exist, use the oldest one.

Before computing GAE on the selected rollout, re-evaluate $V(s_t)$ for all stored states and $V(s_T)$ for its bootstrap state with the *current* critic. Do not modify its stored actions or behavior log-probabilities $\log \pi_{\theta_{\text{old}}}(a_t \mid s_t)$, which define the denominator of r_θ.

Make sure the logged diagnostics are computed on the rollout used for the update.

In your report, describe your implementation and explain why we re-evaluate the values with the current critic, while the stored behavior log-probabilities must not be recomputed. Does re-evaluating the values make the resulting GAE estimate fully on-policy with respect to the current policy? Explain briefly.

(b) **(10 points) Results.** Train PPO with

$$k \in \{0, 2, 4\}.$$

Plot, with one curve per k:

- the evaluation return (`eval_return_mean`);
- the KL divergence between the behavior policy and the current policy at the start of each update (`kl_start`); and
- the clip fraction at the start of each update (`clipfrac_start`).

(c) **(10 points) Analysis.** How stale can the data be before performance degrades? Explain the trends using Q2 and Q4.

The logged clip fraction is

$$\texttt{clipfrac} = \frac{1}{N} \sum_i \mathbf{1}\left[|r_i - 1| > \epsilon\right].$$

Ignoring the rare case of exactly zero advantage,

$$1 - \texttt{clipfrac_start}$$

gives a lower bound on the fraction of samples that contribute a non-zero policy gradient at the start of an update.

Explain why it is only a lower bound rather than the exact fraction.

Use your results to explain why PPO discards its data after every PPO iteration.

Q6 **(30 points) Does the Importance Ratio Help with Stale Data?**

(a) **(8 points) Modification.** Add a choice of surrogate loss. With normalized advantages $\hat{A}$ (as in the provided code), implement the three losses (to be minimized):

`naive` :	$-\hat{\mathbb{E}}\left[\log \pi_\theta(a \mid s)\, \hat{A}\right]$	(no correction)
`ratio` :	$-\hat{\mathbb{E}}\left[r_\theta(s,a)\, \hat{A}\right]$	(importance ratio only)
`clip` :	$-\hat{\mathbb{E}}\left[\min\left(r_\theta \hat{A},\ \mathrm{clip}(r_\theta, 1-\epsilon, 1+\epsilon)\, \hat{A}\right)\right]$	(PPO, provided)

Show that the three losses have the same gradient when

$$r_\theta(s,a) = 1$$

for every sample in the batch.

(b) **(8 points) Results.** Fix $k = 2$ and train with each of the three losses. Plot the evaluation return and the KL divergence between the behavior policy and the policy at the end of each update (`kl_end`).

The $k = 2$, `clip` configuration is already run in Q5 and should be reused.

If a configuration becomes numerically unstable because of the learning objective itself, report when the instability occurs and use the available results up to that point.

(c) **(14 points) Analysis.** Explain the differences. Which failure modes appear without the ratio, and which appear with the ratio but without clipping? Relate your answer to Q2 and Q3. How does clipping help, and what does it *not* fix?